Calculus_Homework 9.11.2026

Zadanie 1: (2.5 pkt)

In Exercises 9–12, v is the velocity of a particle on the line, and $F(t)$ is the position at time t.

11. $v = t^2 + \sqrt{t}$; $F(1) = 1$; find $F(\frac{1}{2})$.

Zadanie 2: (1.5 pkt)

25. Is it true that $\int f(x)g(x)\,dx$ is equal to $[\int f(x)\,dx]\,g(x) + f(x)[\int g(x)\,dx]$?

Zadanie 3: (6.5 pkt)

28. Prove that $\int f(x)f'(x)\,dx = \frac{1}{2}[f(x)]^2 + C$ for any function f.

Zadanie 4: (2.5 pkt)

Find the indefinite integrals in Exercises 29–40.

39. $\int \frac{x^2 + 3}{\sqrt{x}}\,dx$

Zadanie 5: (1.5 pkt)

53. A ball is thrown downward with a velocity of 10 meters per second. How long does it take the ball to fall 150 meters?

Zadanie 6: (3 pkt)

54. A particle moves along a line with velocity $v(t) = \frac{1}{2}t^2 + t$. If it is at $x = 0$ when $t = 0$, find its position as a function of t.

Zadanie 7: (1.5 pkt)

57. A rock is thrown vertically upward with velocity 19.6 meters per second. After how long does it return to the thrower? (The acceleration due to gravity is 9.8 meters per second per second; see Example 4 and Fig. 2.5.3.)

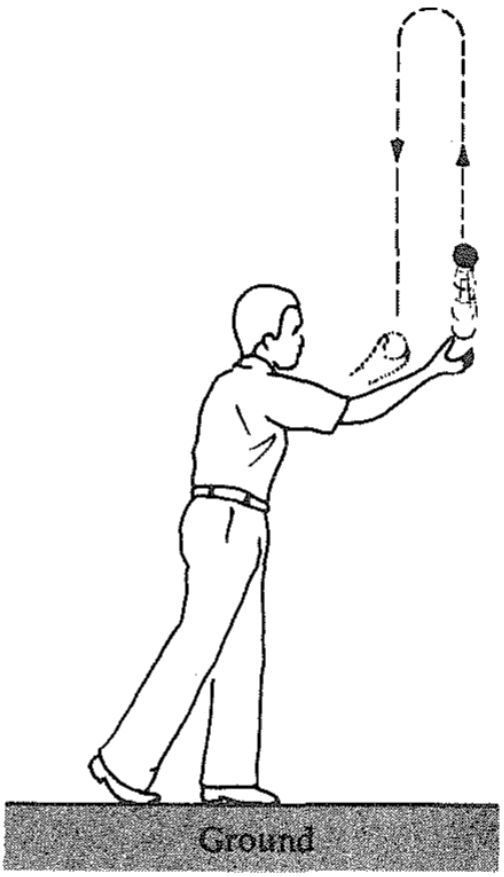


Figure 2.5.3. The path of a rock thrown upwards from the earth.

Zadanie 8: (2.5 pkt)

65. Find the antiderivative $F(x)$ of the function $f(x) = x^3 + 3x^2 + 2$ which satisfies $F(0) = 1$.

Zadanie 9: (2.5 pkt)

68. (a) What integration formula can you derive from the chain rule?

(b) Find $\int[\sqrt{x^2 + 20x} + (x^2 + 20x)](2x + 20)\,dx$.

Zadanie 10: (6.5 pkt)

17. Show that the equation $-s^5 + s^2 = 2s - 6$ has a solution.

Zadanie 11: (6.5 pkt)

19. Prove that $f(x) = x^8 + 3x^4 - 1$ has at least two distinct zeros.

Zadanie 12: (5.5 pkt)

28. An empty bucket with a capacity of 10 liters is placed beneath a faucet. At time $t = 0$, the faucet is turned on, and water flows from the faucet at the rate of 5 liters per minute. Let $V(t)$ be the volume of the water in the bucket at time t. Present a plausible argument showing that V is a continuous function on $(-\infty, \infty)$. Sketch a graph of V. Is V differentiable on $(-\infty, \infty)$?

Zadanie 13: (1.5 pkt)

31. Let $f(x)$ be defined by

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1, \\ ? & \text{if } 1 \leqslant x \leqslant 3, \\ x - 6 & \text{if } 3 < x. \end{cases}$$

How can you define $f(x)$ on the interval $[1, 3]$ in order to make f continuous on $(-\infty, \infty)$? (A geometric argument will suffice.)

Zadanie 14: (3.5 pkt)

37. The function $f(x) = 1/(x - 1)$ never takes the value zero, yet $f(0) = -1$ is negative and $f(2) = 1$ is positive. Why isn't this a counterexample to the intermediate value theorem?